

Who Leaves Teaching, and Why?

The Anatomy of Teacher Attrition in the United States, 1997–2024

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Abstract

Every year about eight in one hundred American teachers leave the profession, and every few years the country rediscovers this number and calls it a crisis. This paper takes the measure the literature already trusts for the level—the March Current Population Survey question comparing last year’s longest job to this week’s—and pushes it as far as it will go: twenty-four supplements spanning calendar 1997–2024, every exit decomposed into its route and destination, priced at the person level by linking consecutive Marches, and modeled with a family margin that existing accounts omit. The anatomy overturns the crisis reading point by point. The rate itself has no trend: 8.3 percent (± 0.2) in 2015–2024, against 7.7 percent in Harris and Adams’s 1992–2001, with Covid a two-year excursion that fully reversed. But leaving teaching is mostly not going to another job: of every 100 leavers, 62 exit the labor force—38 at retirement ages—12 are unemployed, and 26 are working, eight of them still in education and care. For those 26, the move is a lottery, not a raise: the median switcher gains 15 log points while the tenth percentile loses 85, and 47 percent of all leavers report no wage or salary earnings the next year. What predicts exit is the contract, not the person—part-year work, part-time hours, and missing pension coverage dwarf every demographic. The one exception is the family margin: a birth raises a teacher’s exit probability by 4.7 percentage points in the pooled model, and among women the effect, 5.9 points, is double to triple its size in nursing, accounting, or law—though a linked-sample audit shows a third of new-mother leavers are back teaching within a year, so part of this margin is leave rather than loss. Teaching does not lose more people than comparable professions; it loses them differently—to retirement systems and to motherhood—and policy aimed at a generic exodus misses both margins.

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1. Introduction

Teachers are the main school input in the production of student achievement (Rivkin, Hanushek and Kain, 2005; Chetty, Friedman and Rockoff, 2014), and their departures are costly for the students they leave behind (Ronfeldt, Loeb and Wyckoff, 2013). It is therefore no surprise that teacher attrition is among the most measured objects in education policy—or that the measurements have produced three decades of the same argument. Ingersoll (2001) read the Schools and Staffing Survey as evidence of a revolving door; Grissmer and Kirby (1997) read the same numbers as retirements foretold; Harris and Adams (2007) benchmarked teachers against comparable professions in the Current Population Survey and found nothing anomalous; the Learning Policy Institute’s syntheses of the federal school-staffing surveys (Carver-Thomas and Darling-Hammond, 2017; Tan, Wei, Carver-Thomas and García, 2024) returned turnover to the center of the policy agenda; and Aldeman and Yi (2025) recently used the CPS to argue, once more, that turnover is low and stable. The level of teacher attrition may be the best-relitigated statistic in education.

What the literature has not produced is an *anatomy*. The federal surveys count leavers and movers but follow no one out of the school; the CPS studies establish the level and the professional comparison but stop there; the meta-analytic literature on correlates (Nguyen, Pham, Crouch and Springer, 2020) aggregates school-level studies that rarely observe households; and the one administrative study that follows teachers into their next jobs (Brummet, Penner, Pharris-Ciurej and Porter, 2025, in tax records) observes earnings but not households, hours, or comparison professions. Nobody has taken the public instrument this literature already trusts and extracted from it what it actually contains: the routes of exit, the destinations, the earnings consequences, the family triggers, and how all of these have moved over a quarter century measured with a single ruler.

That is this paper. The instrument is the retrospective measure of the March CPS supplement, introduced to this literature by Harris and Adams: a person taught in year t if the occupation of her longest job that year was K–12 classroom teaching, and left the profession if she is no longer teaching in the following March—employed elsewhere, unemployed, or out of the labor force. From the raw Census files I build this measure for every available supplement from survey year 1998 to 2025, recovering the byte layouts of the undocumented early files empirically (Section 2 and Appendix A), which yields an annual attrition series for calendar 1997–2024 and a fully characterized analysis sample—demographics, household structure, earnings, pensions, hours—for 2010–2024. Three things distinguish the exercise from its predecessors. First, *depth*: every exit is decomposed by route and detailed destination; consecutive supplements are linked at the person level to price the move and to audit the timing of family-driven exits; and the exit model includes births and children, constructed from household relationship pointers, for teachers and every comparison profession alike. Second, *time*: each result is presented as a comparison across periods—2010–2014, 2015–2019, 2020–2024, with Harris and Adams’s 1992–2001 estimates as a thirty-year anchor—so that any claim of change confronts the longest constant-instrument series available. Third, *measurement discipline*: I also build the alternative CPS design—the linked monthly panel, twenty entry cohorts, 2005–2024—and audit the two instruments against each other person by person, the first individual-level validation in this literature (Appendix A). Where both designs observe the same stable teacher, they agree on staying or leaving 99.1 percent of the time, and the panel’s apparently higher attrition is fully accounted for by misclassified base years and single-week noise.

The anatomy tells a different story than the exodus debate on either side. The level is flat: 8.3 percent per year in 2015–2024, 9.1 in 2010–2014, 7.7 in 1992–2001; the Covid spike (10.2 percent, measured

in March 2020) had fully reversed by 2021. But the composition is nothing like the image of teachers recruited away by industry. Of every 100 leavers, 62 leave employment altogether—38 at ages 55 and older, the retirement channel; 24 younger, where the family margin operates—12 are unemployed, and 26 hold another job, of whom 8 remain in education and care and only a handful land in the private-sector professional jobs the rhetoric imagines. Linking Marches shows why the outside option disciplines so little: the median employed leaver gains 15 log points of earnings, but the distribution is enormous—the tenth percentile loses 85—and 47 percent of all leavers report no wage or salary earnings the following year. In the exit model, the contract dominates the person: part-year work, part-time hours, and missing pension coverage carry effects an order of magnitude larger than race, marriage, or degrees, a gradient that survives estimator changes, sample cuts, code definitions, and the exclusion of the pandemic (Table 3). The exception is the family margin: a birth raises a teacher’s exit probability by 4.7 percentage points in the pooled model—5.9 among women—versus 2.2 to 3.5 points in the same specification for nursing, accounting, social work, and law. A linked audit tempers and sharpens the finding at once: 35 percent of new-mother leavers are teaching again the following March (against 19 percent of other prime-age leavers), so roughly a third of the margin is parental leave misread as exit—and the remainder, still far above every comparison profession, is real loss. Teaching’s distinction is not how many it loses but *whom* and *when*: mothers at births, veterans at pension thresholds.

The paper proceeds as follows. Section 2 describes data and measurement. Section 3 profiles the workforce at risk. Section 4 establishes the level and its history. Section 5 follows the leavers: routes, destinations, earnings. Section 6 models who leaves, with the robustness battery and the route-specific estimates. Section 7 places teaching among seven comparison professions. Section 8 draws the policy map. Appendix A contains the instrument validation, Appendix B the replication of Harris and Adams on the modern decade, Appendix C the remaining estimation detail.

2. Data and measurement

The data are the public-use person files of the CPS Annual Social and Economic Supplement (ASEC) for survey years 1998–2025, obtained from the Census Bureau’s raw distribution: comma-separated files for surveys 2019–2025, dictionary-parsed fixed-width files for 2011–2018, and, for the earlier supplements, fixed-width files whose byte positions I recover empirically—scanning for the columns that reproduce known identifier joins, age distributions, and occupation code sets, with per-year validation gates—because machine-readable documentation for those vintages no longer exists (Appendix A details the method). Survey year $t+1$ measures attrition for calendar year t ; the assembled series covers calendar 1997–2024 with four gaps (1998 and 2002–2004, whose survey files are either absent from the Census distribution or fail the validation gates), and the fully characterized sample (demographics, household pointers, earnings, pensions, hours, class of worker) is available from survey 2011 onward.

A person is a **teacher** in year t if the occupation of her longest job that year is K–12 classroom teaching: preschool and kindergarten, elementary and middle school, secondary, special education, or teachers not elsewhere classified, under whichever census classification the survey year uses (the 1990 scheme through survey 2002, the 2000/2010 schemes through 2019, the 2018 scheme after); postsecondary teachers and teaching assistants are excluded throughout. The main sample is restricted to college graduates, for Harris and Adams’s original reason: a sixth of CPS “teachers” report no degree, against well under one percent in the NCES universe, and the education-adjacent boundary is exactly where the validation exercise of Appendix A locates the occupation-coding noise. She is a **leaver** if in the following March she is no

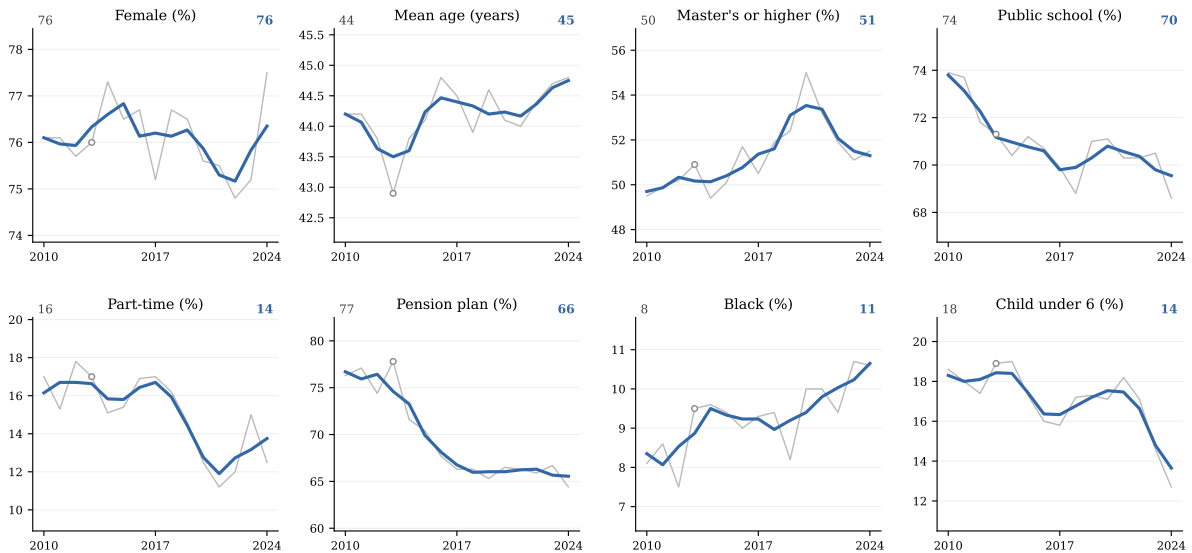


Figure 1: The changing profile of the teaching workforce.

College-graduate K–12 teachers, defined by the longest job of the calendar year; ASEC surveys 2011–2025 (calendar 2010–2024), supplement weights. Thin gray: annual values; blue: three-year centered moving average, with its first and last values labeled above each panel. The hollow 2013 marker uses the reduced five-eighths 2014 file.

longer teaching, with the route read from the labor-force status recode: employed in another occupation, unemployed, or out of the labor force. The status must come from the labor-force recode rather than the occupation field because the unemployed carry their last job’s occupation code; ignoring this detail silently reclassifies displaced teachers as stayers. Own children, including a birth during the year, are attached to every adult through the household parent and spouse pointers. All statistics from survey 2011 onward use supplement weights; sampling errors treat the design with a conservative variance inflation (the published replicate weights are not used), and the earlier series segments are unweighted and marked as such wherever they appear.

Two design limitations discipline the interpretation. The longest-job question is retrospective and often answered by another household member, so it will not register spells shorter than the longest job; and the March week is a single snapshot of the destination. Appendix A quantifies both against the linked panel and shows that, for the questions asked here—profession exit over a year—they are second-order. The converse is not true: the linked panel’s single-week occupation snapshots misclassify enough base-year teachers that its raw attrition rate doubles the retrospective one, entirely through people whose dominant activity that year was not teaching. The measure used here captures leavers only; teachers who change school or district remain teachers, so these rates sit mechanically below the “total turnover” of the school-staffing literature, which adds movers (Ingersoll, 2001; Tan et al., 2024).

3. Who teaches

The workforce against which attrition plays out has been drifting in quiet but consistent directions (Figure 1). Teaching remains about three-quarters female and increasingly credentialed—the share with a master’s degree or more sits above half—while the public-school share has slipped by several points and part-time teaching has edged up. Pension coverage, the profession’s signature amenity and, as Section 6 shows, one of its strongest retention devices, has eroded from 76 to 65 percent across the window. The

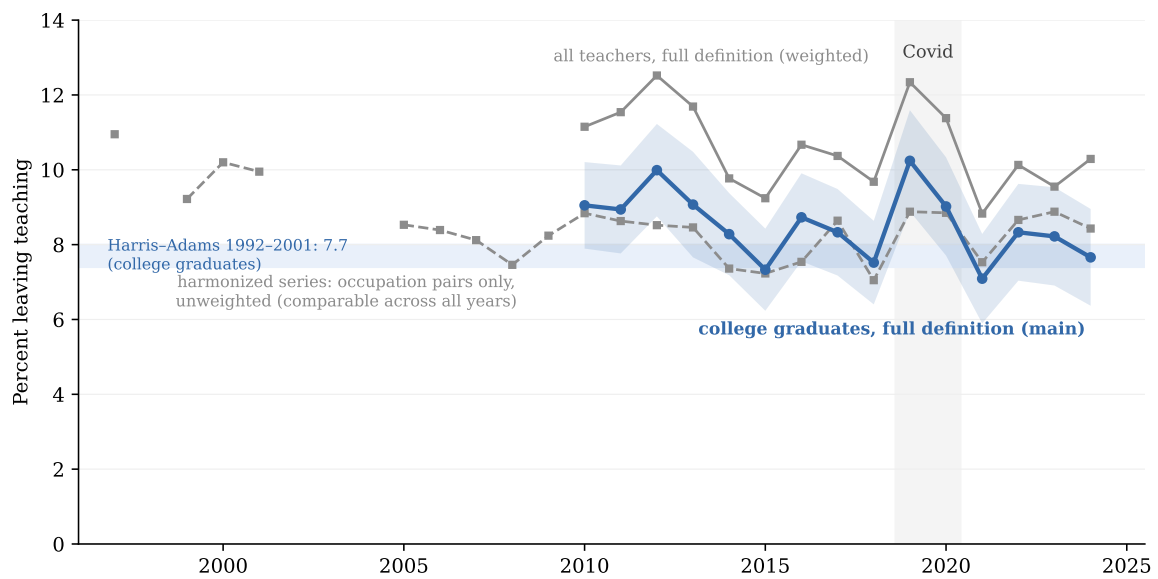


Figure 2: Attrition, 1997–2024, against a thirty-year anchor.

Share of teachers (longest job of the calendar year) not teaching the following March. Blue: college graduates, full route-based definition, weighted, with a 95 percent confidence band. Solid gray: all teachers, full definition, weighted. Dashed gray: the *harmonized* series—occupation pairs only, unweighted, all education levels—the one construction available in every extractable year; its continuity across the 2009/2010 extraction boundary is what validates the early segment. Line breaks mark the years whose files are missing (1998, 2002–2004). Shaded band: Harris and Adams’s (2007) college-graduate estimate for 1992–2001.

Table 1: Attrition by period: level, routes, and sector.

	Leave	To job	Unemp.	Out LF	Public	Private
2010–2014	9.1	2.2	1.1	5.7	8.0	11.8
2015–2019	8.4	2.0	1.1	5.3	7.1	11.5
2020–2024	8.1	2.3	0.9	4.9	6.5	11.7
Harris–Adams 1992–2001	7.7	2.6	0.6	4.5	6.6	–

College-graduate teachers, weighted. Routes sum to the leaver rate. Public/private from the class of worker of the longest job. Harris–Adams row: their Tables 1–2 (routes) and footnote 18 (public), college graduates, March CPS 1992–2001.

age structure is stable: attrition at retirement ages continuously removes the old while entry replenishes the young, so the workforce does not age even though its members do. Mothers of young children are roughly a sixth of the workforce in every year—which is why the family margin estimated below is a first-order object and not a curiosity.

4. How many leave

Pooled over 2015–2024, **8.25 percent** (sampling error ± 0.2) of college-graduate teachers leave the profession per year. Table 1 puts that number in its thirty-year context: 9.1 percent in 2010–2014, 8.4 in 2015–2019, 8.1 in 2020–2024, 7.7 in Harris and Adams’s 1992–2001. If anything, the point estimates drift gently *down* across the modern periods. The internal composition barely moves either: the labor-force exit route carries about five points in every period, the job-switch route two, unemployment one, and the public–private gap is two-to-one in every period, matching the federal follow-up benchmarks (7.9 versus 11.7 percent for 2020–21 in the NCES National Teacher and Principal Survey follow-up). The one

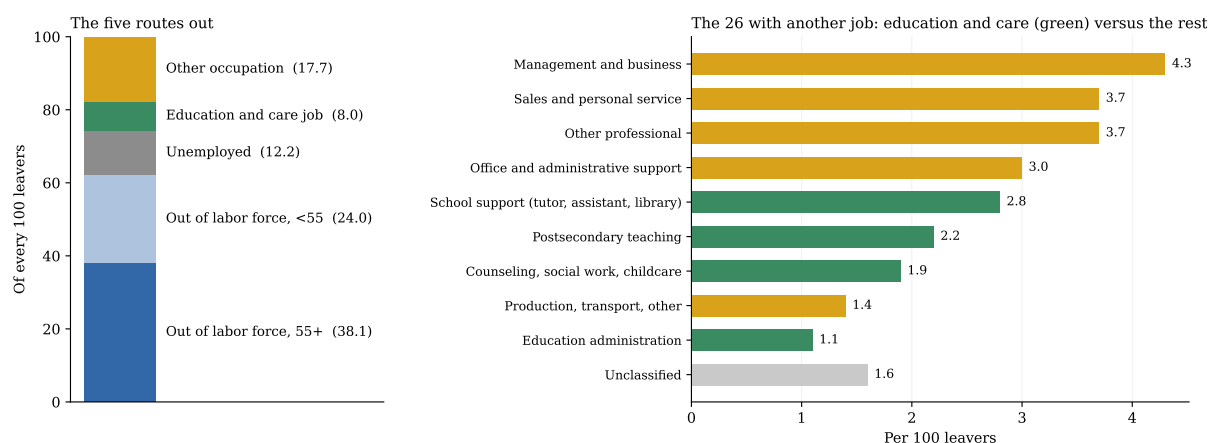


Figure 3: Of every 100 leavers: routes, and the destinations of those who work.

College-graduate teachers pooled 2015–2024, weighted. Both panels use one definition of education and care (green): postsecondary teaching; school support (tutors, assistants, librarians); education administration; and counseling, social work, and childcare. The right panel’s bars sum to the 26 employed leavers of the left panel. Detailed occupations in Appendix C. Sampling errors are roughly ± 1 per component.

genuine excursion is the pandemic: attrition measured in March 2020 reached 10.2 percent, remained elevated in 2021, and was back inside its historical range by calendar 2021—the same reversal [Aldeman and Yi \(2025\)](#) document. Figure 2 adds the long view with the harmonized series (identical construction in every extractable year): the late 1990s sit at 9–11 percent, the 2000s at 7–9, the recent decade at 8–10. Three decades of alarm about accelerating teacher flight are simply not visible in the only long series measured with a constant instrument. What the level conceals, the next two sections open up.

5. Where they go

The composition of exit is the paper’s central fact. Of every 100 leavers, **62 are not employed at all** the following March: 38 have left the labor force at ages 55 or older—the retirement channel—and 24 have left it younger, the margin where childbirth and caregiving operate. Twelve are unemployed. **Twenty-six hold another job**, and their destinations are not where the rhetoric looks for them. Eight of the 26 are still in education and care—postsecondary teaching, tutoring and school support, education administration, counseling and childcare—doing recognizably similar work under a different code. Of the remaining eighteen, management and business absorbs four, office and administrative support three, sales and personal service four, other professional occupations four, and production or transport one. The teacher-turned-tech-worker who anchors so much commentary is, in nationally representative data, a rounding error. In this decomposition the CPS and the tax-record evidence agree ([Brummet et al., 2025](#)): former teachers who work mostly land near education or in general administration, not in highly paid technical fields.

Age organizes the routes with near-mechanical clarity (Figure 4). Job-switching is a young teacher’s exit, peaking in the twenties and declining monotonically thereafter—the early-career pattern the young-graduate studies always found ([Henke and Zahn, 2001](#)); labor-force exit is its mirror image, quiet through mid-career and explosive after 55, the structure [Stinebrickner \(2002\)](#) first documented for teachers and [Grissmer and Kirby \(1997\)](#) read as retirements foretold. Total attrition traces the familiar U, with its trough near 4 percent in the mid-forties: the tenured core of the profession. By 61–64, attrition reaches

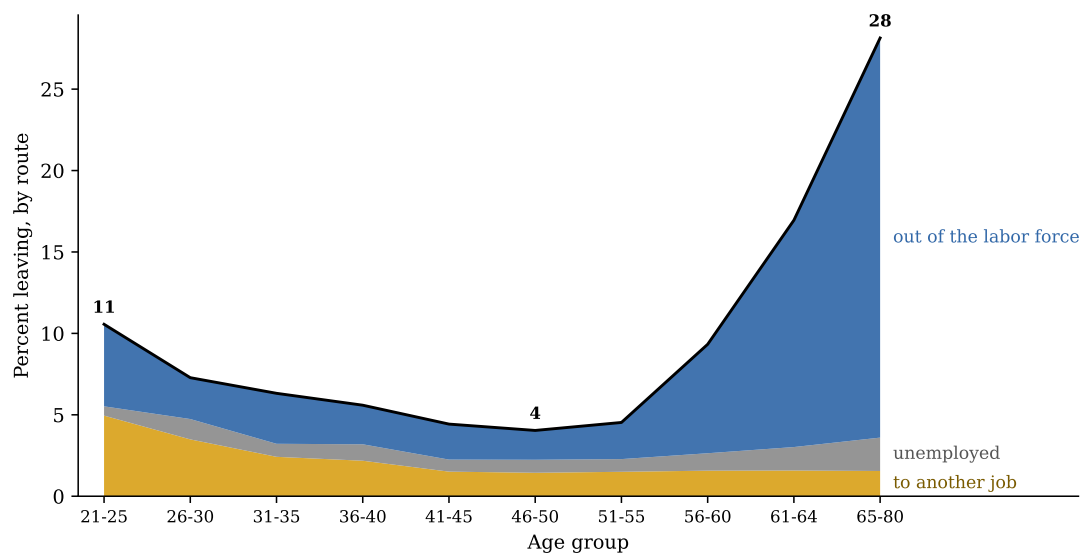


Figure 4: Exit routes over the lifecycle.

Share of teachers leaving by each route, by age group, pooled 2015–2024, weighted. The black line is total attrition; areas stack the routes, labeled in the right margin.

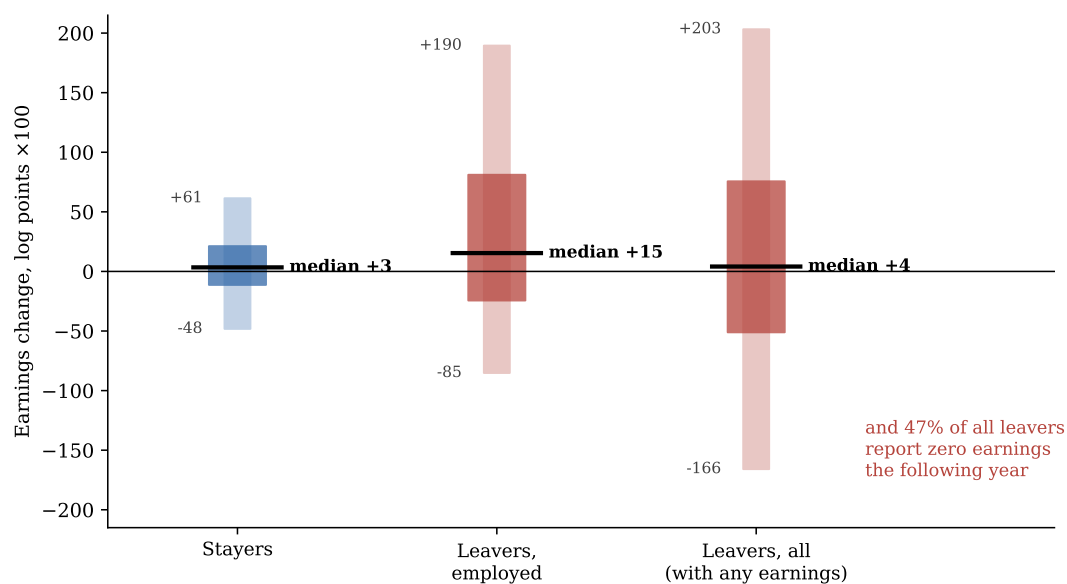


Figure 5: The earnings gamble of leaving.

Change in annual wage and salary earnings (log points $\times 100$) from the last teaching year to the following calendar year, for teachers linked across consecutive supplements by person identifier (validated on sex and age progression), surveys 2011–2025. Boxes: interquartile range; light bars: 10th–90th percentiles. Earnings are proxy-reported and exclude self-employment income, so all distributions carry reporting noise; the comparison across groups is the object of interest.

17 percent, three-quarters of it out of employment altogether. Any aggregate attrition statistic is therefore mostly a statement about the age composition of the workforce, a point the exodus debate routinely ignores.

Linking the same person across consecutive Marches prices the move (Figure 5). Stayers’ median earnings growth is 3 log points; their tails are wide—proxy-reported annual earnings are noisy—which makes the leavers’ distribution the more remarkable, because it is wide *on top of* that baseline noise. Employed leavers’ median gain is 15 log points, but a quarter of them gain more than 81 log points while a tenth

Table 2: Teachers who stay versus teachers who leave.

	Stayers	Leavers	Difference
Age, years	43.9	50.1	+6.2***
Female	76.2%	73.4%	-2.8***
Black	9.5%	10.8%	+1.3*
Married	67.1%	63.7%	-3.3***
Separated or divorced	10.2%	11.7%	+1.5**
Master’s degree or higher	52.4%	45.8%	-6.6***
Public school	71.3%	58.0%	-13.3***
Enrolled in a pension plan	68.0%	51.2%	-16.8***
Part-time (<35 h/week)	11.9%	41.4%	+29.5***
Worked full year (50+ weeks)	75.6%	31.3%	-44.3***
Weekly earnings	\$1,177	\$1,075	\$-102***

Weighted means, college-graduate teachers pooled 2015–2024, measured in the teaching year. Earnings are annual wage and salary divided by weeks worked. */**/**: difference significant at 10/5/1 percent.

lose more than 85. And this is the distribution conditional on earning anything: **47 percent of all leavers report no wage or salary earnings** in the year after leaving (self-employment income, not measured here, is rare among former teachers in tax records; [Brummet et al., 2025](#)). Leaving teaching for work is a gamble that pays for the median mover and costs heavily in the tails; leaving teaching at all is, for nearly half of leavers, a transition into no earned income the CPS can see. The “exit option” that is supposed to discipline teacher pay looks, in the individual data, like a coin flip stapled to a retirement party.

6. Who leaves, and why

Photographed during the last year in the classroom, the leaver differs from the stayer mainly in her contract (Table 2). She is six years older—the retirement tail—somewhat less likely to be in a public school or covered by a pension, and dramatically less attached: 41 percent of leavers taught part-time against 12 percent of stayers, and only 31 percent worked a 50-week year against 76. Race, marriage, and degrees move the comparison by a few points. The teacher who leaves is not, observably, a different kind of person; she is a teacher in a different kind of job-year.

The multivariate ranking (Figure 6) confirms and sharpens the profile, and matches the ordering the meta-analytic literature finds across hundreds of school-level studies ([Nguyen et al., 2020](#)): working conditions first, demographics last. Job attachment dominates: a full 50-week year is associated with 9.3 percentage points less exit—the largest effect in the model by a factor of two—and part-time hours add 4.5 points of risk. The compensation block comes second: pension enrollment retains 2.4 points, public-school employment 1.7, the modern echo of the job-lock Harris and Adams estimated for the 1990s. Demographics are an order of magnitude smaller. The family block would be too—children as such barely matter—except for its first year: **a birth during the year raises exit probability by 4.7 points** in the pooled model, roughly half of baseline attrition, conditional on age, education, marriage, and existing children; estimated on women alone the effect is 5.9 points (Section 7 compares professions).

Because a birth and an exit are measured in the same window, timing deserves its own audit. A child under one in March $t+1$ may have arrived after the teaching spell ended, and a teacher on leave in the March week is recorded out of the labor force even if she returns. Linking mothers forward one more

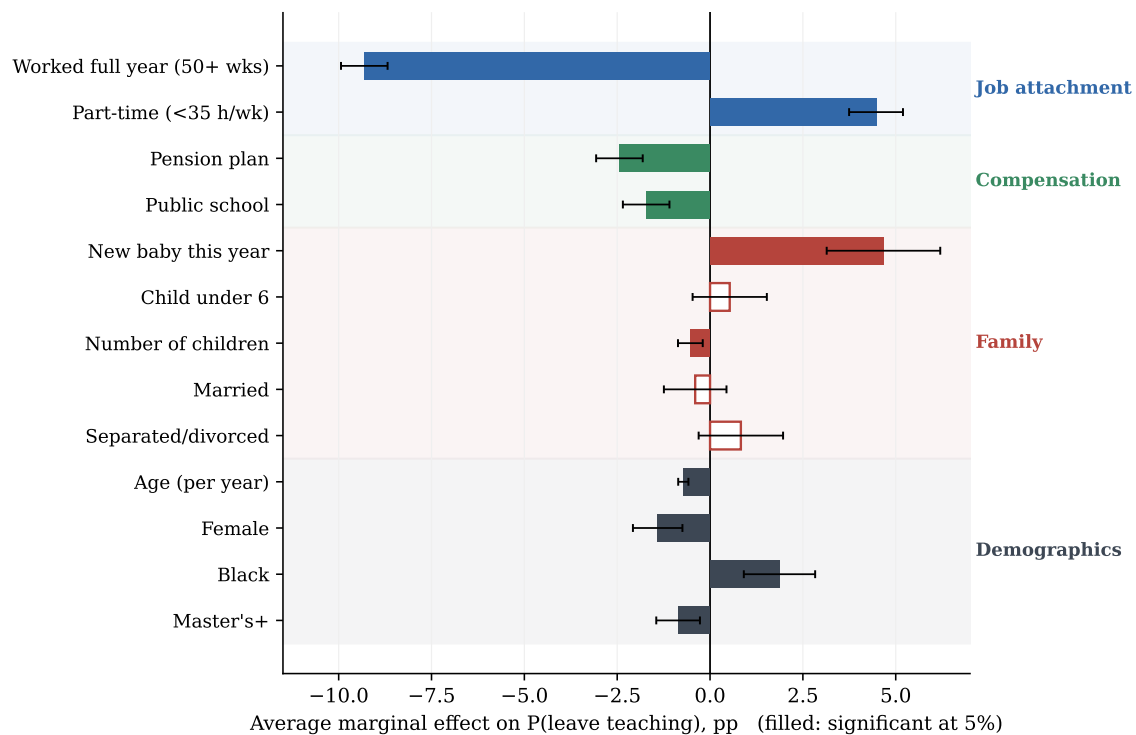


Figure 6: What predicts leaving.

Average marginal effects from a probit of leaving on characteristics of the teaching year, survey-year fixed effects, college-graduate teachers pooled 2015–2024 ($n = 28,932$). Whiskers: 95 percent confidence intervals; filled bars significant at 5 percent, outlined bars not. Background bands group the covariates. Age enters quadratically; its full profile is Figure 4.

Table 3: Robustness of the key margins.

	Full year	Part-time	Pension	New baby	Public	<i>n</i>
Baseline probit	-9.31* (0.32)	4.47* (0.37)	-2.44* (0.32)	4.67* (0.78)	-1.72* (0.32)	28,932
Weighted LPM	-12.55* (0.54)	8.42* (0.85)	-2.53* (0.45)	5.52* (1.23)	-1.48* (0.47)	28,932
Age under 55	-7.63* (0.55)	8.41* (0.95)	-2.60* (0.45)	5.67* (1.25)	-2.92* (0.49)	22,683
Public school only	-10.31* (0.59)	11.21* (1.21)	-2.98* (0.54)	4.80* (1.30)	—	20,665
Women only	-12.42* (0.60)	8.02* (0.93)	-1.91* (0.50)	5.79* (1.49)	-1.03 (0.53)	22,136
Excluding Covid surveys	-9.18* (0.35)	3.96* (0.40)	-2.32* (0.35)	4.85* (0.87)	-1.47* (0.36)	23,301
2010-2014 window	-10.68* (0.44)	4.69* (0.49)	-3.19* (0.46)	4.21* (0.97)	-1.01* (0.45)	16,364
Core codes only (2300-2330)	-8.79* (0.33)	4.10* (0.39)	-2.51* (0.32)	4.37* (0.78)	-0.82* (0.34)	26,134

Each row re-estimates the exit model under the stated change; cells report the marginal effect in pp with standard errors in parentheses, * significant at 5 percent. Baseline: probit AMEs; subsample rows use the weighted linear probability model. “Core codes”: teacher defined by 2300–2330 only. The Public column is degenerate within the public-only sample.

year bounds this: **35 percent of new-mother leavers are teaching again the following March**, against 19 percent of other prime-age leavers. Roughly a third of the new-baby margin is therefore parental leave misread as exit—and the remaining two-thirds, still twice any comparison profession’s total effect, is genuine loss. Both numbers matter for policy: the first says schools recover part of these exits within a year; the second says the recovery is far from complete.

Table 3 stress-tests the result. The attachment gradient—full-year retaining, part-time expelling—survives the switch to a linear probability model, the exclusion of the pandemic surveys, the earlier 2010–2014 window, and the strict core-code definition of teaching; the pension and new-baby effects are similarly

Table 4: Routes are chosen by different variables.

	To another job	Unemployed	Out of labor force
Worked full year (50+ wks)	-0.79* (0.19)	-1.32* (0.14)	-7.32* (0.27)
Part-time (<35 h/wk)	1.06* (0.23)	0.54* (0.14)	2.53* (0.27)
Pension plan	-0.42* (0.18)	-0.50* (0.12)	-1.65* (0.25)
New baby this year	-0.12 (0.46)	-0.32 (0.44)	4.92* (0.58)
Child under 6	-0.06 (0.26)	-0.31 (0.21)	0.92* (0.43)
Female	-0.92* (0.19)	-0.19 (0.13)	-0.17 (0.27)
Age (per year)	-0.18* (0.05)	-0.00 (0.03)	-0.31* (0.06)
Public school	-1.46* (0.18)	-0.25* (0.12)	0.05 (0.26)
Master's+	-0.14 (0.17)	-0.15 (0.12)	-0.63* (0.23)

Probit AMEs (pp) estimated separately for each exit route, same specification as Figure 6; robust standard errors in parentheses, * significant at 5 percent. Route effects need not sum exactly to the single-equation total.

Table 5: The top-risk decile.

	Top-risk decile	All teachers
Actual leaver rate %	33.0	8.2
Age	54.4	44.4
Female %	74.4	76.0
Part-time %	75.0	14.3
Full-year %	4.3	71.9
Pension %	35.6	66.6
Public %	45.2	70.2

Teachers in the top decile of predicted exit probability from the model of Figure 6, versus all teachers; the realized attrition rate in the first row validates the ranking. Descriptive, in-sample.

stable. Restricting to ages under 55 shrinks the full-year coefficient by roughly two-fifths, exactly as it should if part of it proxies phased retirement, while leaving the part-time, pension, and baby margins intact; the within-public-school and within-women estimates reproduce the same ordering. Nothing in the battery disturbs the reading that contract shape, deferred compensation, and childbirth are the three levers.

Estimating the model route by route (Table 4) shows that the aggregate coefficients blend three distinct decisions. The labor-force route absorbs most of the full-year effect (-7.3 of -9.3) and essentially all of the new-baby effect ($+4.9$ of the $+4.7$ pooled total): mothers who leave, leave employment, not for other jobs. The job-switch route is where youth and gender operate—men and the young switch more—and the unemployment route, small everywhere, is the only one that moves with the cycle. A retention policy aimed at “leavers” in the aggregate is thus aiming at three different populations: retirees, new mothers, and loosely attached part-timers, in that order of size.

7. Teaching among the professions

Measured with the same ruler, teaching’s attrition is the opposite of anomalous (Figure 7). Teachers (8.3 percent) sit at the registered-nurse benchmark (8.0), below physical therapists (10.4) and pharmacists (10.6), five points below social workers (13.6), above accountants (6.3) and lawyers (4.8)—and every

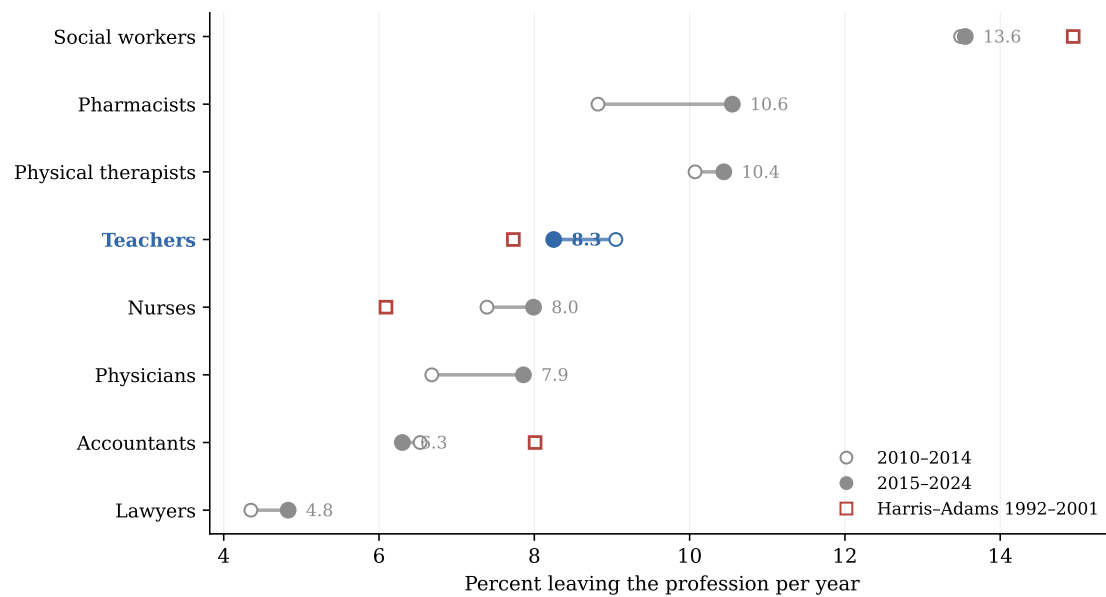


Figure 7: Leaving the profession: eight professions, three vintages.

Annual profession-leaving rates under the identical instrument, college graduates, weighted: open circles 2010–2014, filled circles 2015–2024, red squares Harris and Adams’s 1992–2001 estimates where available. Profession defined by the longest job of the calendar year; leaving includes unemployment and labor-force exit.

one of these positions is essentially unchanged from the first half of the decade, as most are essentially unchanged from Harris and Adams’s 1990s. What is distinctive about teaching is compositional. Its labor-force exit share (62 percent of attrition) far exceeds nursing’s (44) and social work’s (21); its pension coefficient is two to three times any comparison profession’s (Appendix B); and its family margin is an outlier (Figure 8): the new-baby effect for female teachers (+5.9 points) doubles nursing’s (+2.6) and law’s (+2.4, the latter imprecise), and exceeds accounting’s (+3.5); the effects for social work, physicians, and therapists are too imprecise to distinguish from zero. Nursing—the profession most like teaching in education, feminization, and caretaking—loses new mothers at half the rate teaching does, a comparison that should focus attention on what differs between the two jobs: schedule rigidity within the school year, and the thinness of leave provisions relative to the health sector.

8. What it means for policy

The anatomy replaces one large question—why are teachers leaving?—with three small ones, each with its own answer. *Why do most leavers leave?* Because they retire: 38 of every 100 exits happen out of the labor force at 55-plus, on a schedule written into pension rules (Podgursky, 2003). This is the predictable, plannable part of attrition, and treating it as crisis distorts everything downstream. *Who leaves before retirement?* Disproportionately, teachers whose contracts never attached them—part-year, part-time, pension-less positions carry many times the exit risk of any demographic group—and new mothers, whose exit effect has no counterpart of that size in any comparison profession even after netting out the third who return within a year. These are the two actionable margins, and they point at specific instruments: converting part-year attachments into full-year contracts, extending pension coverage where it is thin (disproportionately private schools, whose attrition doubles the public sector’s in every period), and building leave and within-year flexibility that lets a teacher have a child without leaving employment—the

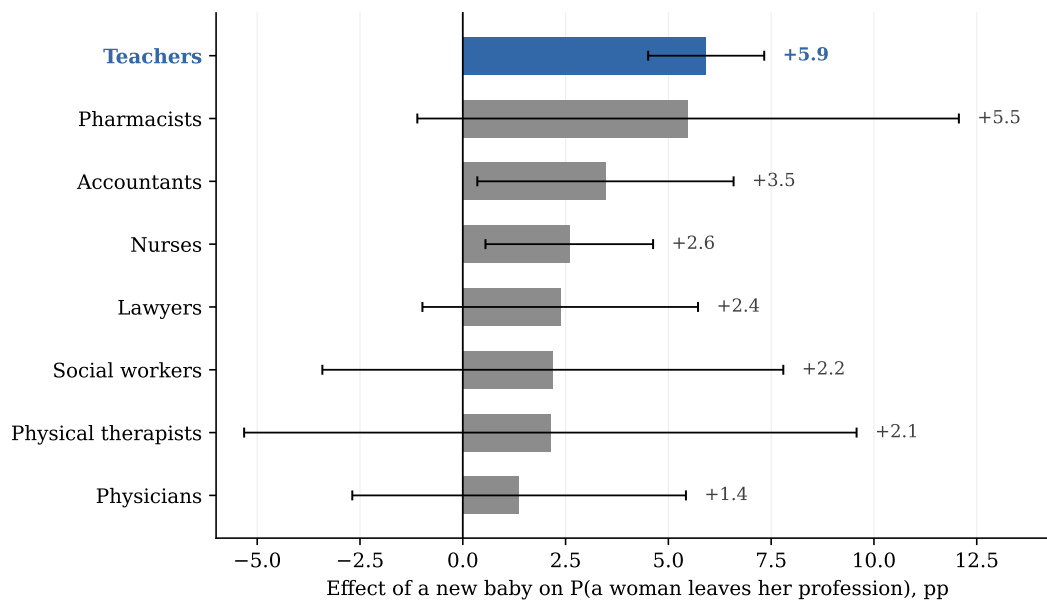


Figure 8: A new baby and the probability that a woman leaves her profession.

Average marginal effect of a child under one, from profession-specific probits on women (age quadratic, marriage, advanced degree, other children), pooled 2010–2024. Whiskers: 95 percent confidence intervals; small-profession estimates are imprecise. A linked-sample audit (Section 6) indicates roughly a third of the teacher effect reflects leave rather than permanent exit.

margin where nursing demonstrably outperforms teaching. *And who is being poached?* Almost no one: 26 of 100 leavers work at all the next year, eight of those inside education and care, and the earnings distribution of the movers is a lottery, not a market verdict against teacher pay. The outside option constrains teacher retention far less than the inside conditions do—which is, on balance, good news for the policymakers who control the inside conditions.

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Appendix A. Validating the instrument person by person

The CPS admits two designs for measuring teacher attrition, and they disagree by a factor of two. This appendix explains the disagreement at the level of individual people and shows why the retrospective design is the right one for profession exit.

The alternative design links individuals across the CPS rotation scheme: a household is interviewed in four consecutive months, rests eight, and is interviewed in the same four calendar months a year later (Madrian and Lefgren, 2000). From the raw monthly files I built the twenty entry cohorts of 2005–2024, kept the unambiguous teachers—employed in K–12 teaching in *all four* initial interviews—and defined a panel-leaver as one who teaches in *none* of the four follow-ups. For the subsample whose initial window includes March, the same person also answered the next ASEC’s retrospective questions, so the two instruments can be cross-tabulated person by person. (The identifiers that join the supplement to the monthly files predate their published documentation in several vintages; their byte positions were recovered by scanning for the offset at which the 22-character person identifier reproduces the March monthly identifiers—a join that cannot succeed by accident—and the same empirical approach, anchored on occupation code sets, age distributions, and identifier joins with validation gates, recovers the undocumented 1998–2010 supplement layouts used for the long series of Figure 2.)

Table 6: Panel classification versus March retrospective, twenty cohorts pooled.

Linked-panel classification	March retrospective classification			
	Not a teacher	Stayed	Left	Total
Stayer (teaches all 4 post months)	198	7,911	67	8,176
Leaver (teaches in none)	1,018	10	257	1,285
Total	1,216	7,921	324	9,461

Teachers stable across all four initial monthly interviews, observed in all four follow-ups, window including March, linked to the following ASEC ($n = 9,461$). “Not a teacher”: the retrospective codes the longest job of the base year outside K–12 teaching.

Where the two instruments agree the person was a base-year teacher, they almost never disagree about what happened next: **99.1 percent agreement**, 77 true conflicts in 9,461 cases, 98–100 percent in every cohort (Table 6). The entire gap between the panel’s apparent attrition (13.6 percent among these stable teachers) and the retrospective’s (3.9) sits in the 13 percent of panel-teachers whom the retrospective does not count as teachers at all—and their longest jobs concentrate in teaching assistance, other-instruction, education administration, postsecondary teaching, and childcare, with 84 percent of them panel-“leavers.” In words: the panel’s excess attrition is within-education reshuffling and single-week noise (summer non-employment, temporary absences), not profession exit. The panel measures leaving the *classroom this week*; the retrospective measures leaving the *profession this year*. For this paper the second is the estimand—and the widely quoted point-in-time panel rates of 15 percent and above should not be read as profession exit.

Appendix B. Harris and Adams (2007) on the modern decade

Their design, applied verbatim to 2015–2024, returns their world with interpretable drift (Table 7): teaching within half a point of its 1990s level; nursing converged up to teaching; accounting drifted below;

Table 7: Leaving the profession, by type: 1992–2001 versus 2015–2024.

	Harris & Adams, 1992–2001				This paper, 2015–2024			
	All	Switch	Unemp.	Out LF	All	Switch	Unemp.	Out LF
Teachers	7.73	2.59	0.61	4.53	8.25	2.12	1.00	5.12
Nurses	6.09	1.68	0.86	3.54	7.99	3.66	0.81	3.53
Social workers	14.94	10.87	1.34	2.73	13.55	9.45	1.22	2.89
Accountants	8.01	4.10	1.55	2.36	6.30	1.79	1.18	3.32

Annual rates in percent, college graduates, March CPS. 1992–2001 from Harris and Adams (2007), Tables 1–2; 2015–2024 computed here on ASEC 2016–2025. “Switch”: employed in another occupation; “Out LF”: out of the labor force in the survey week.

social work still far above. In the regression counterpart, the teacher–nurse gap ranges from +0.3 to –1.8 points across control sets, while social workers exit five to six points more than observably similar teachers. Teaching’s two structural signatures survive intact: the labor-force route carries about 60 percent of its attrition in both eras, and the pension coefficient (–3.6 points) is two to three times any comparison profession’s. The question this literature has argued over since Ingersoll—is teacher attrition exceptional?—has had the same answer for thirty years.

Appendix C. Additional estimation output

Table 8: Annual attrition series, calendar 1997–2024.

Calendar year	College graduates	All teachers	Old-style (unwtd.)	Obs.
1997	—	—	10.9%	2,620
1998	—	—	—	—
1999	—	—	9.2%	2,743
2000	—	—	10.2%	2,646
2001	—	—	9.9%	4,632
2002	—	—	—	—
2003	—	—	—	—
2004	—	—	—	—
2005	—	—	8.5%	4,161
2006	—	—	8.4%	4,336
2007	—	—	8.1%	4,357
2008	—	—	7.5%	4,448
2009	—	—	8.2%	4,357
2010	9.1% (0.6)	11.2%	8.8%	4,301
2011	8.9% (0.6)	11.5%	8.6%	4,184
2012	10.0% (0.6)	12.5%	8.5%	4,192
2013	9.1% (0.7)	11.7%	8.5%	2,896
2014	8.3% (0.6)	9.8%	7.4%	4,333
2015	7.3% (0.6)	9.2%	7.2%	3,972
2016	8.7% (0.6)	10.7%	7.5%	4,045
2017	8.3% (0.6)	10.4%	8.6%	4,053
2018	7.5% (0.6)	9.7%	7.0%	3,986
2019	10.2% (0.7)	12.3%	8.9%	3,321
2020	9.0% (0.7)	11.4%	8.8%	3,232
2021	7.1% (0.6)	8.8%	7.5%	3,040
2022	8.3% (0.7)	10.1%	8.7%	3,141
2023	8.2% (0.7)	9.6%	8.9%	3,006
2024	7.7% (0.7)	10.3%	8.4%	2,918

College-graduate (with sampling errors in parentheses), all-teacher, and harmonized old-style definitions. Rows shown as dashes are the gaps: survey files for 1999, 2003, and 2004 are absent from the Census distribution and the 2005 file fails the validation gates. Segments before calendar 2010 are unweighted, all-education, occupation pairs only. The 2013 row uses the reduced five-eighths 2014 file.

Table 9: Determinants of leaving teaching: full probit.

	AME (pp)	SE
Age (per year)	-0.72	(0.07)
Age squared	0.010	(0.000)
Female	-1.41	(0.34)
Black	1.87	(0.49)
Married	-0.40	(0.43)
Separated or divorced	0.83	(0.58)
Master's degree or higher	-0.86	(0.30)
Public school	-1.72	(0.32)
Enrolled in a pension plan	-2.44	(0.32)
Part-time (<35 h/week)	4.47	(0.37)
Worked full year (50+ weeks)	-9.31	(0.32)
Number of children	-0.53	(0.17)
Child under 6	0.53	(0.51)
New baby during the year	4.67	(0.78)

Average marginal effects, percentage points; college-graduate teachers pooled 2015–2024, survey-year fixed effects. Children variables built from household parent and spouse pointers.

Table 10: Detailed destination occupations of switchers.

Destination occupation	Share of switchers
Postsecondary teachers	8.5%
Managers, all other	6.0%
Teaching assistants	4.9%
Education and childcare administrators	4.2%
Lawyers	3.7%
Tutors	2.6%
Childcare workers	2.3%
Customer service representatives	2.1%
Secretaries and administrative assistants, except legal, medical, and executive	1.7%
Retail salespersons	1.3%

Detailed March occupations among leavers employed at $t+1$, pooled 2015–2024, duplicate categories across census coding schemes merged; the grouped version is Figure 3, right panel.